

Physics Week 9 Text-Only Pilot Note

Baseline vs candidate v2, text-only, DeepSeek dev and held-out splits

Reading guide. Candidate v2 improved over the baseline on both the development and held-out text-only splits, but severe-error rate did not improve on held-out. Treat this as a Physics Week 9 pilot result, not a cross-course or multimodal conclusion.

Experiment State

Dev validation

8/8 + 8/8

baseline and candidate passed

Held-out exact

84.3%

candidate v2 test split

Held-out total MAE

2.08

candidate v2, points

Severe error

44.4%

no held-out improvement

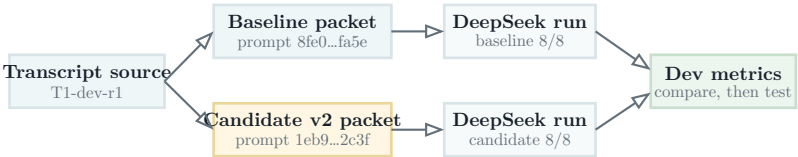
Protocol decision. Candidate v2 was frozen before held-out evaluation. The real held-out runs passed 18/18 and metrics are recorded in HELD-OUT-METRICS-STRICT-SCHEMA.md.

Data boundary. Student data and model outputs remain under ignored **Data/**. This Git record stores paths, hashes, commands, and aggregate metrics only.

Reproducibility Anchors

Anchor	Value
Record directory	experiments/records/physics-week9-baseline-candidate-v2-run
Branch	codex/physics-week9-baseline-candidate-v2-run
Model-run commit	95622f0aead87187c6410ec0a38ba94cb2866dee
Report input records	DEV-METRICS-STRICT-SCHEMA.md; HELD-OUT-METRICS-STRICT-SCHEMA.md
Course / assessment	physics / week9
Splits	development, 8 students; held-out, 18 students
Provider / model	deepseek / deepseek-v4-pro
Endpoint	https://api.deepseek.com
Input mode	text-only, pilot-derived automatic transcript
Temperature / response format	0 / json_object
Typst	0.15.0

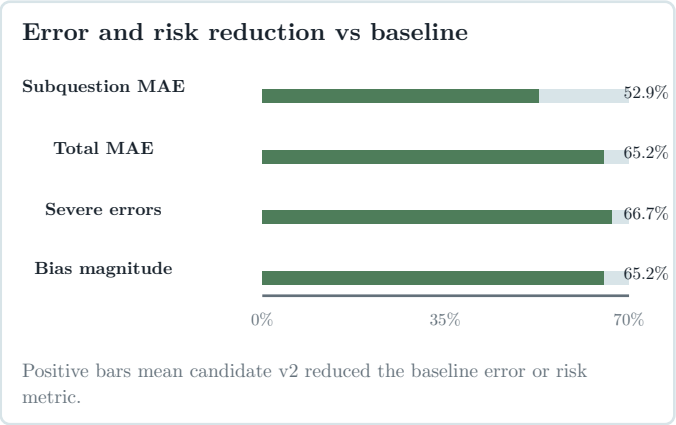
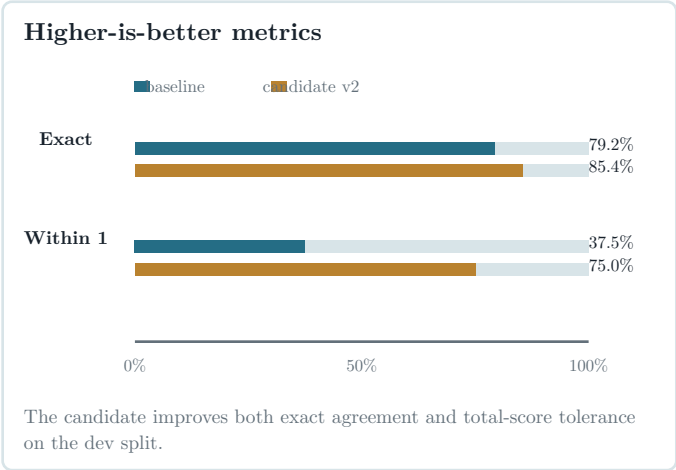
Prompt Packet Lineage



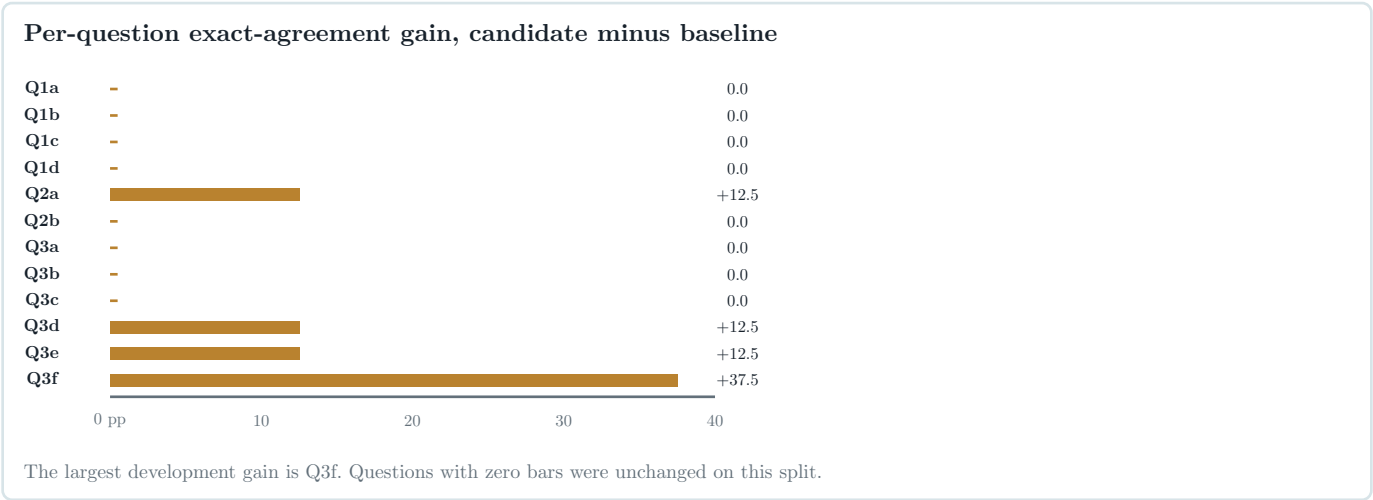
Both packets use the same transcript hash 30e45836b26f6c05d0a55c2e436d5ace7078d01ae86932c3c64b27ad14e24cf8 and rubric hash a02e0531b3d78590c66d32e76eba170d2e0400e3e4a1c60436f4f5c2c8e93b21.

Development Results

Metric	Baseline	Candidate v2	Direction
exact agreement	0.7917	0.8542	candidate better
macro accuracy	0.7917	0.8542	candidate better
subquestion MAE	0.2214	0.1042	candidate better
total score MAE	2.1563	0.7500	candidate better
within 1 point rate	0.3750	0.7500	candidate better
severe error rate	0.3750	0.1250	candidate better
mean signed error	-0.1797	-0.0625	candidate less biased



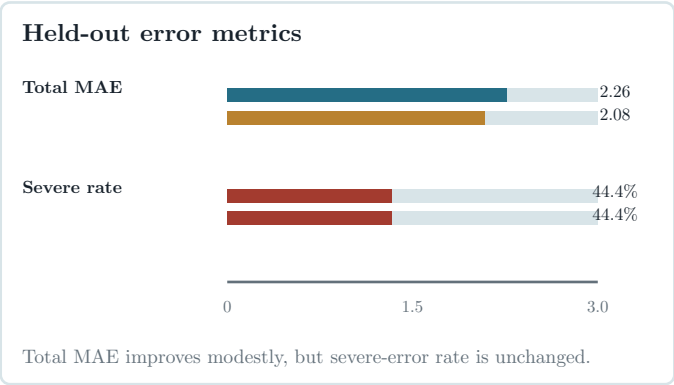
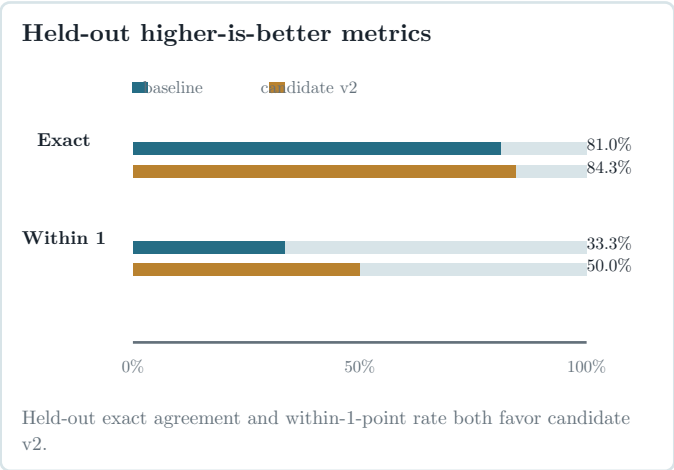
Where Candidate v2 Changed The Most



Held-Out Test Results

Held-out interpretation. Candidate v2 is better than the baseline on most aggregate held-out metrics, but severe-error rate is unchanged. The result supports candidate v2 as the current text-only pilot winner, while severe-error reduction remains the next improvement target.

Metric	Baseline	Candidate v2	Direction
exact agreement	0.8102	0.8426	candidate better
macro accuracy	0.8102	0.8426	candidate better
subquestion MAE	0.2002	0.1852	candidate better
total score MAE	2.2639	2.0833	candidate better
within 1 point rate	0.3333	0.5000	candidate better
severe error rate	0.4444	0.4444	no improvement
mean signed error	−0.1539	−0.1389	candidate slightly less biased



Held-out per-question changes

Improved	Q1c +16.7 pp	Q3e +11.1 pp	Q3f +11.1 pp	Q2a/Q2b +5.6 pp
Declined	Q1d −5.6 pp	Q3b −5.6 pp		
Unchanged	Q1a	Q1b	Q3a/Q3c/Q3d	

The largest held-out gain is Q1c. The severe-error caveat means total-error cases need manual review before operational claims.

Prompt And Packet Registry

This PDF uses short hashes for layout. Full SHA-256 values are recorded in DEV-METRICS-STRICT-SCHEMA.md and HELD-OUT-METRICS-STRICT-SCHEMA.md.

Role	Prompt source	Prompt hash
baseline	experiments/records/.../prompts/grade_standard_v1_strict_schema.txt	8fe0...fa5e
candidate v2	experiments/records/.../prompts/grade_candidate_v2_strict_schema.txt	1eb9...2c3f

Role	Packet path	Packet hash
baseline	Data/physics/benchmark/text_packets/physics-week9-baseline-text-strict-schema/G1-dev-r1	e556...2601
candidate v2	Data/physics/benchmark/text_packets/physics-week9-candidate-v2-text-strict-schema/G1-dev-r1	8987...ed6d

Development Run Outputs Used For Metrics

Role	Run directory	Validation
baseline	Data/physics/benchmark/runs/physics-week9-baseline-candidate-v2/deepseek-baseline-text-G1-dev-r1-strict-schema	passed, 8/8
candidate v2	Data/physics/benchmark/runs/physics-week9-baseline-candidate-v2/deepseek-candidate-text-G1-dev-r1-strict-schema	passed, 8/8

Local artifact	Hash
dev-metrics-strict-schema.json	68d4...e613
dev-student-total-errors-strict-schema.csv	3e11...b176

Held-Out Run Outputs Used For Metrics

Role	Run directory	Validation
baseline	Data/physics/benchmark/runs/physics-week9-baseline-candidate-v2/deepseek-baseline-text-G1-test-r1-strict-schema	passed, 18/18
candidate v2	Data/physics/benchmark/runs/physics-week9-baseline-candidate-v2/deepseek-candidate-text-G1-test-r1-strict-schema	passed, 18/18

Local artifact	Hash
held-out-metrics-strict-schema.json	a360...4de9
held-out-student-total-errors-strict-schema.csv	3ff3...bbc7

Reproduction Commands

Run from the repository root. The API key must be supplied through DEEPSEEK_API_KEY; never write the key value into Git, reports, or shell history.

The exact multiline commands are stored in STRICT-SCHEMA-RERUN.md and HELD-OUT-PREFLIGHT.md; the user-facing DeepSeek key setup instructions are stored in EXTERNAL-API-DATA-GATE.md.

Freeze Gate Before Held-Out Test

Gate	Status	Evidence
Development validation	passed	both strict-schema runs validated 8/8
Prompt packet record	passed	prompt, packet, text-source, and rubric hashes are recorded
Candidate v2 freeze	passed	CANDIDATE-V2-FREEZE.md records the frozen prompt
English rubric rule	passed	model-facing prompts and rubrics are English
External API data gate	passed	anonymous data approved for DeepSeek public API by supervisor
Held-out packets	passed	HELD-OUT-PREFLIGHT.md records 18/18 dry-runs
Held-out real API run	passed	baseline and candidate both validated 18/18
Held-out metrics	recorded	HELD-OUT-METRICS-STRICT-SCHEMA.md

Limitations

Development split has 8 anonymous students; held-out split has 18. This is text-only and does not test multimodal image grading. The text source is a pilot-derived automatic transcript, and the gold reference is single-primary-rater rather than adjudicated. Failed baseline attempts are retained under ignored Data/ for audit but are not used for metrics. Held-out severe-error rate did not improve, so severe-error reduction remains unresolved. This note supports the Physics Week 9 text-only pilot result, not a final cross-course conclusion.